

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Other
Manufacturer: Bosch
Firmware Version: H/W 1.03, FPGA 4.00.3959, FW04.61.424
Model(s): DCN-CCU2

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0002-b001	1.4.2

Version History

Module Version	Date	Notes
1_15_0_0	6/6/2018	- Removed model DCN-CCUB2 - Added DualDelegateMicrophoneControl - Removed DualDelegateControl - Removed DualDelegateControlLastActioned - Added values to WirelessMicrophoneControl command to match MicrophoneControl command.
1_11_0_0	4/16/2018	Added CCU parameter to Microphone Control.
1_9_0_0	1/23/2018	Increased number of supported microphones.

Global Scripter Module Communication Sheet

1_7_0_0	10/23/2017	Added commands: - DualDelegateControl - DualDelegateControlLastActioned - WirelessMicrophoneControl - WirelessMicrophoneControlLastActioned MicrophoneControlLastActioned
1_0_0_0	1/16/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- For ID assignment, the device will automatically assign IDs to the Microphone units connected to the main unit. Chairman Microphone units will take precedence over Normal Microphone units. For example, if there are 2 chairman microphones, those will take Microphone 1 and 2 within the module. The rest of the Microphone units will take Microphone 3 and after.
- `InterfaceName.Update('HeartBeat')` **must be called every 5 seconds** to keep the connection alive to the device. If no reply from the client after 3 heartbeat commands (15 secs), the DCN-CCU2 will disconnect the session.

Supported Classe and Example

EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9451, Model='DCN-CCU2')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command ClearRequestToSpeakListAndSpeakers List	Value 'String'		
# ClearRequestToSpeakListAndSpeakersList example InterfaceName.Set('ClearRequestToSpeakListAndSpeakersList', 'String')			
Command DualDelegateMicrophoneControl ^{1, 2}	Value 'On'	Value 'Off'	Value 'Add to Speakers List'
	'Remove from Speakers List'	'Add to Request To Speak List'	'Remove from Request To Speak List'
Qualifier Key 'Dual Delegate Unit'	Qualifier Value '1' – '245'		
Qualifier Key 'L/R'	Qualifier Value 'Left'	Qualifier Value 'Right'	
Qualifier Key 'CCU'	Qualifier Value '1' – '30'		
# DualDelegateMicrophoneControl example InterfaceName.Set('DualDelegateMicrophoneControl', 'On', {'Dual Delegate Unit': '1', 'L/R': 'Left', 'CCU': '1'})			
Command MicrophoneControl ^{1, 2}	Value 'On'	Value 'Off'	Value 'Add to Speakers List'
	'Remove from Speakers List'	'Add to Request To Speak List'	'Remove from Request To Speak List'
Qualifier Key 'Microphone'	Qualifier Value '1' – '245'		
Qualifier Key 'CCU'	Qualifier Value '1' – '30'		
# MicrophoneControl example InterfaceName.Set('MicrophoneControl', 'On', {'Microphone': '1', 'CCU': '1'})			
Command WirelessMicrophoneControl ¹	Value 'On'	Value 'Off'	Value

	'Add to Speakers List'
'Remove from Speakers List'	'Add to Request To Speak List'
	'Remove from Request To Speak List'
Qualifier Key 'Microphone'	Qualifier Value '1' – '245'
# WirelessMicrophoneControl example InterfaceName.Set('WirelessMicrophoneControl', 'On', {'Microphone': '1'})	

¹ The permitted state transitions and the state transitions initiated by the microphone button depend on the Microphone Mode setting on the device and the permitted number of Speakers allowed.

'On' will turn On the mic and add it to the Speakers List if there are vacancies.

'Off' will turn Off the mic and remove it from the Speakers List and the Request to Speak List.

'Add to Speakers List' will add the mic to the Speakers List if there are vacancies.

'Remove from Speakers List' will remove the mic to the Speakers List if it is in the Speakers List

'Add to Request To Speak List' will add the mic to the Request To Speak List

'Remove from Request To Speak List' will remove the mic to the Request To Speak List if it is in the Request To Speak List.

² When the CCU parameter is extended above 1, only 'On' and 'Off' are supported. In addition, the status for 'Microphone Control Last Actioned', 'Request to Speak List' and 'Speakers List' are not supported.

Status Available

For RequestToSpeakList and SpeakersList, call Update to receive the latest status. ConnectionStatus, DualDelegateMicrophoneControl, MicrophoneControl, MicrophoneControlLastActioned, WirelessMicrophoneControl, and WirelessMicrophoneControlLastActioned do not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	
DualDelegateMicrophoneControl	'On'	'Off'	
Qualifier Key	Qualifier Value		
'Dual Delegate Unit'	'1' – '245'		
Qualifier Key	Qualifier Value	Qualifier Value	
'L/R'	'Left'	'Right'	
Qualifier Key	Qualifier Value		
'CCU'	'1' – '30'		
# DualDelegateMicrophoneControl examples Value = InterfaceName.ReadStatus('DualDelegateMicrophoneControl', {'Dual Delegate Unit': '1', 'L/R': 'Left', 'CCU': '1'}) InterfaceName.SubscribeStatus('DualDelegateMicrophoneControl', None, FeedbackHandler)			
Command	Value	Value	Value
MicrophoneControl ¹	'On'	'Off'	'Request To Speak'
Qualifier Key	Qualifier Value		
'Microphone'	'1' – '245'		
Qualifier Key	Qualifier Value		
'CCU'	'1' – '30'		
# MicrophoneControl examples Value = InterfaceName.ReadStatus('MicrophoneControl', {'Microphone': '1', 'CCU': '1'}) InterfaceName.SubscribeStatus('MicrophoneControl', None, FeedbackHandler)			

Command MicrophoneControlLastActioned ²	Value '1' – '245'
# MicrophoneControlLastActioned examples Value = InterfaceName.ReadStatus('MicrophoneControlLastActioned') InterfaceName.SubscribeStatus('MicrophoneControlLastActioned', None, FeedbackHandler)	
Command RequestToSpeakList ³	Value '0' – '245'
Qualifier Key 'Index'	Qualifier Value '1' – '25'
# RequestToSpeakList examples InterfaceName.Update('RequestToSpeakList') Value = InterfaceName.ReadStatus('RequestToSpeakList', {'Index': '1'}) InterfaceName.SubscribeStatus('RequestToSpeakList', None, FeedbackHandler)	
Command SpeakersList ³	Value '0' – '245'
Qualifier Key 'Index'	Qualifier Value '1' – '25'
# SpeakersList examples InterfaceName.Update('SpeakersList') Value = InterfaceName.ReadStatus('SpeakersList', {'Index': '1'}) InterfaceName.SubscribeStatus('SpeakersList', None, FeedbackHandler)	
Command WirelessMicrophoneControl ¹	Value 'On' Value 'Off'
Qualifier Key 'Microphone'	Qualifier Value '1' – '245'
# WirelessMicrophoneControl examples Value = InterfaceName.ReadStatus('WirelessMicrophoneControl', {'Microphone': '1'}) InterfaceName.SubscribeStatus('WirelessMicrophoneControl', None, FeedbackHandler)	
Command WirelessMicrophoneControlLastActioned	Value '1' – '245'
# WirelessMicrophoneControlLastActioned examples Value = InterfaceName.ReadStatus('WirelessMicrophoneControlLastActioned') InterfaceName.SubscribeStatus('WirelessMicrophoneControlLastActioned', None, FeedbackHandler)	

¹ The permitted state transitions and the state transitions initiated by the microphone button depend on the Microphone Mode setting on the device and the permitted number of Speakers allowed.

'On' means the mic is in the Speakers List.

'Request To Speak' means the mic is in the Request to Speak List.

'Off' means the mic is in neither the Speakers List nor the Request To Speak List.

² For Dual Delegate Microphones, the Dual Delegate Unit number will be recorded.

³ Qualifier parameters is not required when calling the Update method with this command because the response returns statuses for all qualifier states.

Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	9451
Logon Credentials Supported:	No
Multi-Connection	Yes
Capabilities:	
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

[illegible]

[illegible]

Global Scripter Module
Communication Sheet

Revision: 6/6/2018

Dual Delegate Microphone Control Remove from Request To Speak List Dual Delegate Unit 245 L/R Right CCU 30	\x03'\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\xF5=\xFF\xFF
Dual Delegate Microphone Control Remove from Speakers List Dual Delegate Unit 245 L/R Right CCU 30	\x031\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\xF5=
Microphone Control Add to Request To Speak List Microphone 1 CCU 1	\x03>\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01\x02\xFF\xFF
Microphone Control Add to Speakers List Microphone 1 CCU 1	\x030\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01\x02
Microphone Control Off Microphone 1 CCU 1	\x03"\x00C\x15\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01\x02\x00\x031\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01\x02\x03'\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01\x02\xFF\xFF
Microphone Control On Microphone 1 CCU 1	\x03"\x00C\x15\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01\x02\x01
Microphone Control Remove from Request To Speak List Microphone 1 CCU 1	\x03'\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01\x02\xFF\xFF
Microphone Control Remove from Speakers List Microphone 1 CCU 1	\x031\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01\x02
Microphone Control Add to Request To Speak List Microphone 1 CCU 30	\x03>\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01<\xFF\xFF
Microphone Control Add to Speakers List Microphone 1 CCU 30	\x030\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01<
Microphone Control Off Microphone 1 CCU 30	\x03"\x00C\x15\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01<\x00\x031\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01<\x03'\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01<\xFF\xFF
Microphone Control On Microphone 1 CCU 30	\x03"\x00C\x15\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01<\x01
Microphone Control Remove from Request To Speak List Microphone 1 CCU 30	\x03'\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01<\xFF\xFF
Microphone Control Remove from Speakers List Microphone 1 CCU 30	\x031\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\x01<
Microphone Control Add to Request To Speak List Microphone 245 CCU 1	\x03>\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\xF5\x02\xFF\xFF
Microphone Control Add to Speakers List Microphone 245 CCU 1	\x030\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\xF5\x02
Microphone Control Off Microphone 245 CCU 1	\x03"\x00C\x15\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\xF5\x02\x00\x031\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\xF5\x02\x03'\x00C\x16\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x00\xF5\x02\xFF\xFF

[illegible]

Wireless Microphone Control Remove from Speakers List Microphone 245	\x031\x00C\x14\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x01\x00\xF5>
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Appendix B. Update Commands

HeartBeat	'pD\x00\x10\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00
Request To Speak List Index 1	\x03%\x00C\x12\x00\x00\x00\x00\x00\x00\x00\x00\x00
Request To Speak List Index 25	\x03%\x00C\x12\x00\x00\x00\x00\x00\x00\x00\x00\x00
Speakers List Index 25	\x03.\x00C\x12\x00\x00\x00\x00\x00\x00\x00\x00\x00